[Def.
On an interval I, the differential equation of the form

$$L(y) = a_n \cdot y^{(n)}(t) + a_{n-1} \cdot y^{(n-1)}(t) + \cdots + a_1 y'(t) + a_0 y(t) = b(t)$$

is called the linear equation. If $b(t) \equiv 0$, we say that homogeneous.

[Observation.
The L is a differential operator on $C^{(n)}$, because: for any $c \in \mathbb{R}$ and $y_1, y_2 \in C^{(n)}$,

$$L(cy_1 + y_2) = \sum_{i=0}^{n} a_i \cdot (cy_1 + y_2)^{(i)}(t) = \sum_{i=0}^{n} \left[c \cdot a_i y_1^{(i)}(t) + a_i y_2^{(i)}(t) \right] = c \cdot L(y_1) + L(y_2).$$

Order of 2.

Thm. Let $a, b \in \mathbb{R}$ be constants, and consider the linear equation

$$L(y) = y'' + a_1 y' + a_2 y = 0.$$

If r_1, r_2 are distinct root of the characteristic polynomial

$$p(r) = r^2 + a_1 r + a_2,$$

then

$$\phi(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} \quad (c_1, c_2 \in \mathbb{R})$$

is the general solution for $L(y) = 0$.

If r_1 is a repeated root of p, then

$$\phi(x) = c_1 e^{r_1 x} + c_2 x e^{r_2 x} \quad (c_1, c_2 \in \mathbb{R})$$

is the general solution.

Prof. observe that

$$L(e^{rx}) = r^2 \cdot e^{rx} + a_1 r e^{rx} + a_2 e^{rx} = (r^2 + a_1 r + a_2) \cdot e^{rx} = p(r) \cdot e^{rx}$$

and

$$L(x e^{rx}) = (x e^{rx})'' + a_1 (x e^{rx})' + a_2 (x e^{rx})$$

$$= (r e^{rx} + r e^{rx} + r^2 x e^{rx}) + a_1 \cdot e^{rx} + a_1 r \cdot x e^{rx} + a_2 x e^{rx}$$

$$= (r^2 + a_1 r + a_2) \cdot x e^{rx} + (2r + a_1) e^{rx}$$

$$= (p(r) \cdot x + p'(r)) \cdot e^{rx}$$

$p(r) = p'(r) = 0 \iff r$ is multiple root of p, so Proved.

Non-Homogeneous form.

Thm. If e_1, e_2 are two solution of

$$y'' + a_1 y' + a_2 y = 0 \quad \text{on an interval } I \text{ containing } x_0.$$

then $W(e_1, e_2)(x) = e^{-a_1(x-x_0)} \cdot W(e_1, e_2)(x_0)$.

Proof. we have $\begin{cases} e_1'' + a_1 e_1' + a_2 e_1 = 0 \\ e_2'' + a_1 e_2' + a_2 e_2 = 0 \end{cases}$, so multiplying $-e_2$ and e_1 at first and second, respectively.

and adding: $\begin{cases} -e_1'' e_2 - a_1 e_1' e_2 - a_2 e_1 e_2 = 0 \\ e_1 e_2'' + a_1 e_1 e_2' + a_2 e_1 e_2 = 0 \end{cases}$

$$= e_1 e_2'' - e_1'' e_2 + a_1(e_1 e_2' - e_1' e_2) = 0 \quad \cdots (*)$$

Notice that $(e_1 e_2' - e_1' e_2)' = e_1 e_2'' + e_1' e_2' - e_1' e_2' - e_1'' e_2 = e_1 e_2'' - e_1'' e_2$

Therefore the $(*)$ is: for $W = W(e_1, e_2)$,

$$W' + a_1 W = 0 \Rightarrow W = C \cdot e^{-a_1 x}$$

Now, $C = W(x_0) \cdot e^{a_1 x_0}$, we obtain the result. ∎

Consider that:

$$L(y) = y'' + a_1 y' + a_2 y = b(x)$$

If ψ, ψ_p are solution of $L(y) = b(x)$, then

$L(\psi - \psi_p) = b - b = 0$, so $\psi - \psi_p$ is a solution of $L(y) = 0$.

Therefore, $\psi - \psi_p = C_1 e_1 + C_2 e_2$ where e_1, e_2 are solution of $L(y) = 0$

So, if we a one 'particular' solution ψ_p, then we get the general solution.

Particular solution.

Let $y_p(x) = u_1(x)\cdot y_1(x) + u_2(x)\cdot y_1(x)$. Then,

$$b = y_p'' + a_1 y_p' + a_2 y_p$$

$$= u_1'' y_1 + 2u_1' y_1' + u_1 y_1'' + a_1 u_1' y_1 + a_1 u_1 y_1' + a_2 u_1 y_1$$
$$+ u_2'' y_2 + 2u_2' y_2' + u_2 y_2'' + a_1 u_2' y_2 + a_1 u_2 y_2' + a_2 u_2 y_2$$
$$= \underbrace{u_1 \cdot L(y_1) + u_2 \cdot L(y_2)}_{=0}$$
$$+ (u_1'' y_1 + u_2'' y_2) + 2(u_1' y_1' + u_2' y_2') + a_1(u_1' y_1 + u_2' y_2)$$

Now, if $u_1' y_1 + u_2' y_2 = 0$, then $u_1'' y_1 + u_2'' y_2 + u_1' y_1' + u_2' y_2' = 0$,
So, only $u_1' y_1' + u_2' y_2' = b$. is remained.

Let's solve the equation $\begin{cases} u_1' \cdot y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = b. \end{cases}$

Multiplying y_2' to $u_1' y_1 + u_2' y_2 = 0$, then we obtain

$u_1' y_1 y_2' + u_2' y_2 y_2' = 0$. Substituting $u_2' y_2' = b - u_1' y_1'$.

Then we obtain $u_1' y_1 y_2' + y_2 b - u_1' y_1' y_2$

$= u_1' \cdot W(y_1, y_2) = -y_2 b.$

Therefore, proceed to similar process to u_2', we obtain:

$$u_1' = \frac{-y_2 b}{W(y_1, y_2)} \quad \text{and} \quad u_2' = \frac{y_1 \cdot b}{W(y_1, y_2)}.$$

Consequently,

$$y_p(x) = \int_{x_0}^{x} b(t) \cdot \begin{vmatrix} y_1(t) & y_1(x) \\ y_2(t) & y_2(x) \end{vmatrix} \cdot \frac{1}{W(y_1, y_2)(t)}\, dt.$$